Engineering Portfolio Project

Finite Element Analysis of a Right-Angle Bracket

With and Without Stiffener

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Mechanical Engineering Portfolio

Contents

1. Study Objective

1.1 Objectives

The objective of this study is to assess the behaviour of a constrained right-angle bracket submitted to a load. It has to be verified that the yield strength is not reached anywhere in the part.

1.2 Analysis Type

A static analysis has been performed, considering that our part is fixed on a wall, and has to withstand a uniform pressure of 24,53 kPa on its upper face.

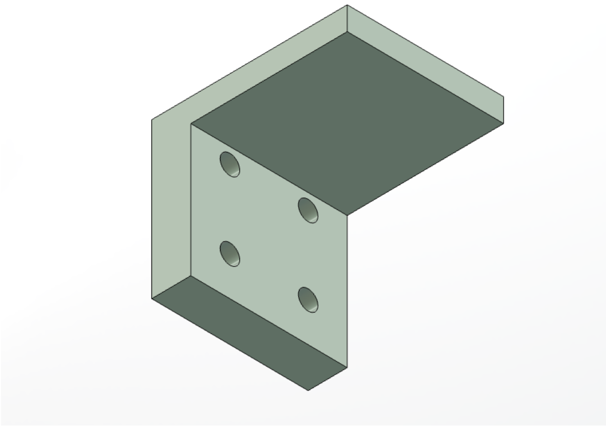
1.3 Methodology

For this analysis two different cases have been compared: the bracket without stiffener on one hand, and the bracket with stiffener on the other hand.

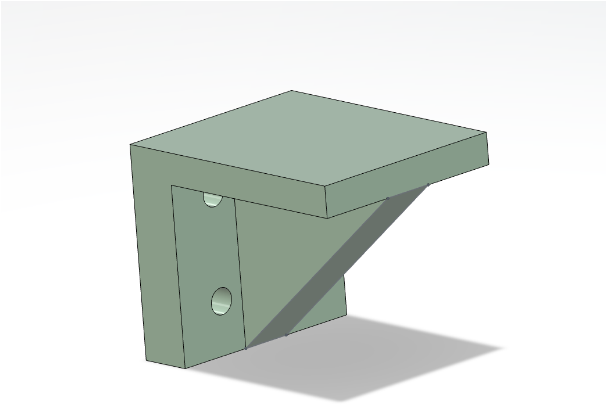
2. Geometrical Assumptions

The parts in this study has been modelled via the 3DExperience platform, and then imported in the FREECAD software through .step files.

2.1 Geometry Overview



Bracket without stiffener



Bracket with stiffener

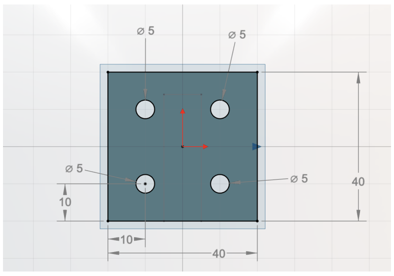
2.2 Unit System

The geometry is given in "mm", the unit system « mm-N-tonne-MPa-mJ » has been thus selected.

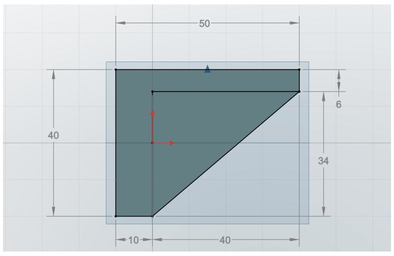
2.3 Characteristic Dimensions

The characteristic dimensions are given in the figures below:

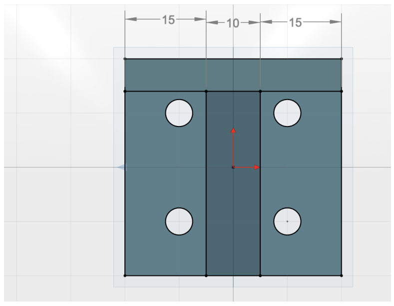
- Holes have a diameter of 5 mm
- The face in contact with the wall is a square of 40 mm of side length
- The upper (lifting) face of the bracket is a rectangle whose dimensions are 40 mm and 50 mm.
- The stiffener has the shape of an extruded triangle, with the right faces having lenghts of 49 mm and 34 mm, and the extrusion length being 10 mm.



Rear view



Side view



Front view

2.4 Problem Symmetries

The geometry and the loading assumptions being symmetric, one symmetry plane could have been used. However this symmetry has not been used during this study.

2.5 Geometrical Modelling Space

A 3D modelling has been used.

3. Physical Behaviour Assumptions

3.1 Material Description

The selected material is Steel-15CrNi6, whose parameters are displayed on figure 1. We can find the values for the following parameters:

- Density
- Young'Modulus
- Poisson ratio
- Thermal conductivity
- Expansion coefficient
- Specific Heat Capacity

Added to this parameter, the tensile strength of the used material has been found in [1], and has a value of 610 MPa.

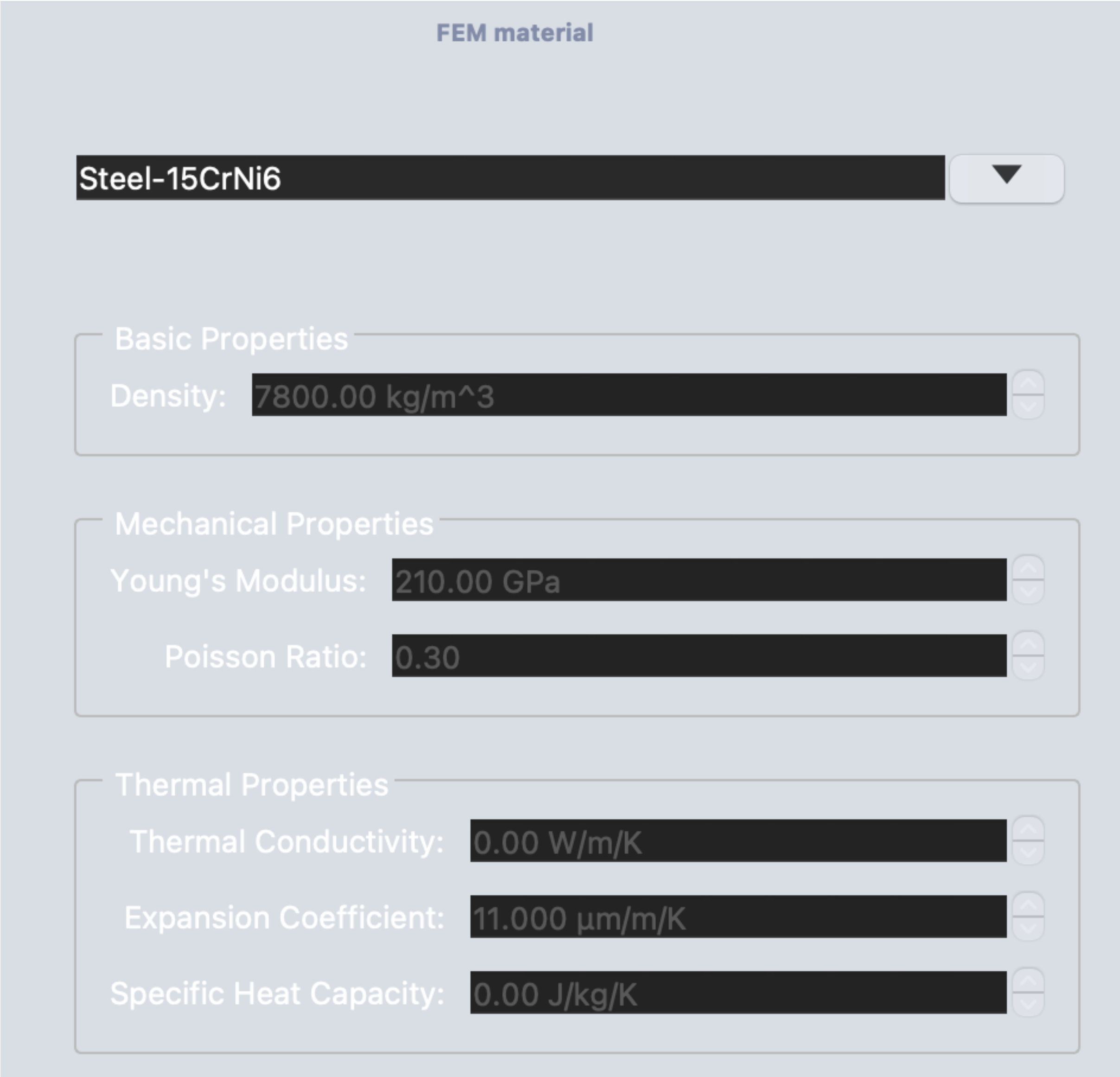


Figure 1: Material properties

3.1.1 Mechanical Behaviour Model

The mechanical behaviour of steel is considered as being elastic linear and isotropic.

3.2 Section Definition

Section type solid, homogeneous, material: Steel-15CrNi6.

4. Loading Assumptions

4.1 Boundary Conditions Model

The four holes in the parts were modeled to represent the faces in contact with the 4 screws holding the parts on the wall. These areas were thus fixed, and a constant stress has been applied on the upper face of our bracket. We made sure that on the fixed surfaces, we had 0 degrees of freedom. Regarding the uniform stress, here are the assumptions that we made:

$$A = L * l \tag{1}$$

$$F_m = m * g \tag{2}$$

$$p = F_m/A \tag{3}$$

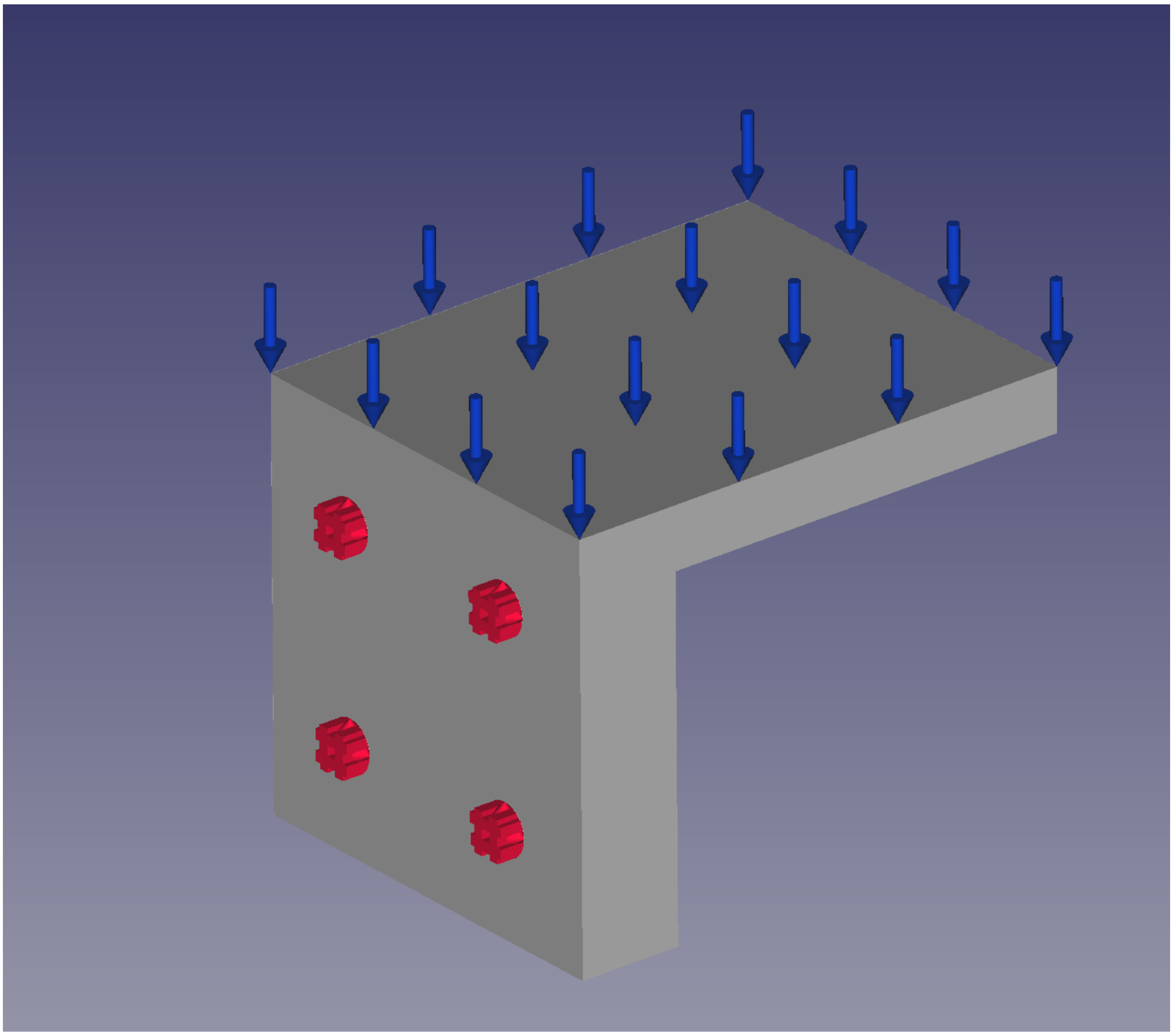
where:

- A is the area of the lifting surface
- L & l are the two side lengths of the lifting surface
- m is the mass of the object
- g is the acceleration due to gravity
- F_m the force applied by this object on the surface
- p is the calculated stress

On table 1, the values for these parameters can be found.

Table 1: Used parameters and their value

Parameter	**Value**
l [mm]	40
L [mm]	50
A [mm^2]	2000
m [kg]	5
g [m/s^2]	9,81
F_m [N]	49,05
p [Pa]	24525



Applied boundary conditions

4.2 Rigid Body Modes

The cylindrical faces were completely constrained as mentioned previously, so no degree of freedom left, and thus, no rigid body modes.

4.3 Coordinate Systems

For the analysis, the global coordinate system has been used.

4.4 Spatial and Temporal Distribution

There is no spatial or temporal distribution regarding the loading or the boundary conditions.

4.5 Initial Conditions

No initial condition, as only a static analysis has been performed.

5. Discretization and Mesh Assumptions

5.1 Finite Element Type Selection

3D elements have been chosen for the parts.
Elements family: free tetrahedral elements.
Quadaratic shape functions.

5.2 Meshing Method

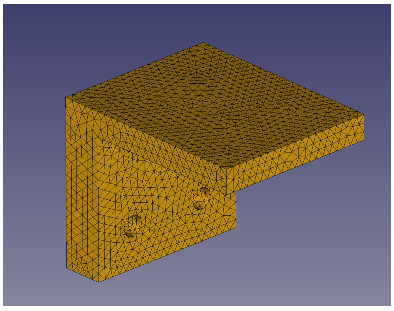
To create the parts meshing, the tool "FEMMeshNetgen" has been used on FREECAD.

5.3 Mesh Size and Number of Elements

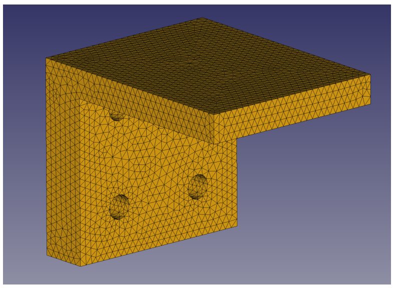
Table 2: Number of elements for each mesh

Type of mesh	Coarse Mesh	Medium Mesh	Fine Mesh
Without stiffener	17366	47082	93434
With stiffener	20379	59616	117735

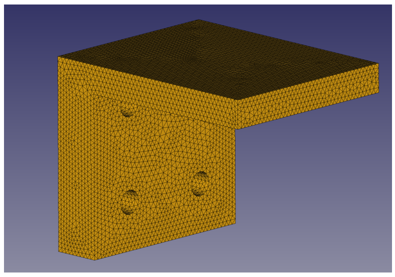
5.4 Optimized Mesh



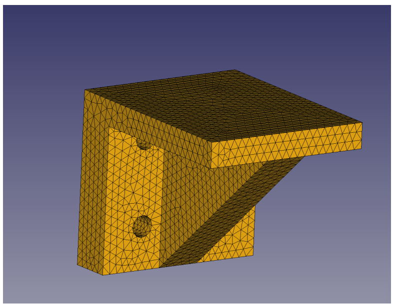
Without stiffener – Coarse mesh



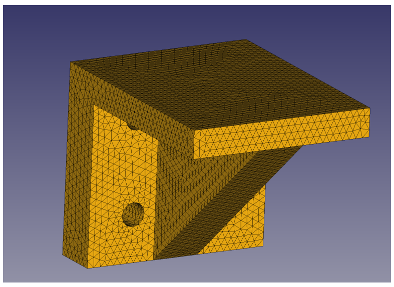
Without stiffener – Medium mesh



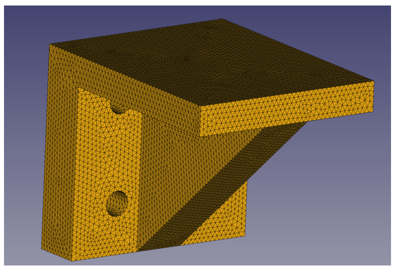
Without stiffener – Fine mesh



With stiffener – Coarse mesh



With stiffener – Medium mesh



With stiffener – Fine mesh

Figure 2: Meshes used for the convergence study. Top row: bracket without stiffener. Bottom row: bracket with stiffener.

5.5 Mesh Convergence Study

In order to perform the convergence study, it has been decided to use three meshes:

- one coarse mesh

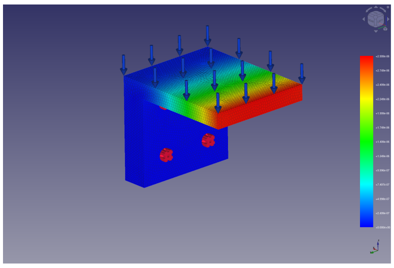
- one medium mesh
- one refined mesh

6. Results

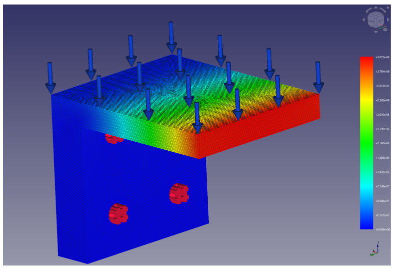
Two types of results were computed: the displacement magnitude, and the Von Mises stress. For each of the three meshes, a screenshot of these results has been taken, and is displayed below. The solver used on FREECAD is "Solver CalculiX Standard".

6.1 Configuration A: Bracket Without Stiffener

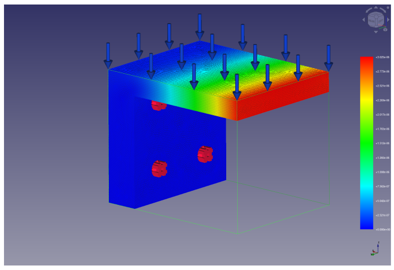
6.1.1 Displacement Results



Coarse mesh



Middle mesh



Refined mesh

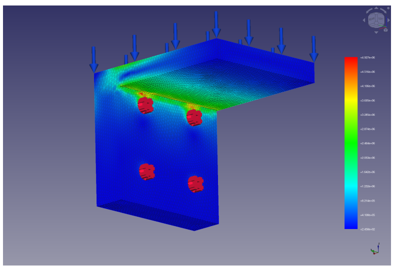
Table 3: Max. displacement magnitude for each mesh

Type of mesh	Coarse Mesh	Medium Mesh	Fine Mesh
Displacement magnitude [nm]	3000	3020	3020

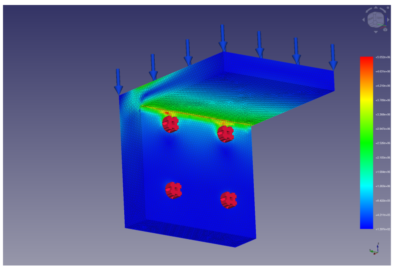
On the figures above, it can be seen that the maximum deflection is reached at the tip of the lifting face, as forecast by the beam theory.

As can be seen in table 3, the displacement magnitude seems to have converged at 3020 nm, as this value remained constant between the medium mesh and the refined mesh.

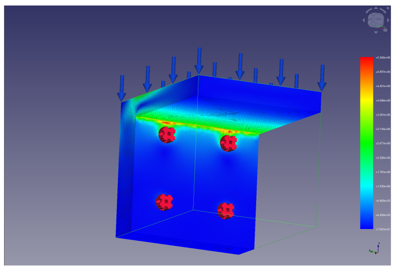
6.1.2 Von Mises Stress Results



Coarse mesh



Middle mesh



Refined mesh

Table 4: Max. Von Mises stress for each mesh

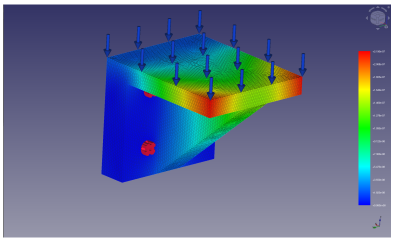
Type of mesh	Coarse Mesh	Medium Mesh	Fine Mesh
Von Mises stress [kPa]	1264,8	1214,76	1412,7

As can be seen on the figures above, the maximum Von Mises stress is reached at the corner joining the two plates. This result was also expected, as the stress is generally the highest in the corner for this loading case.

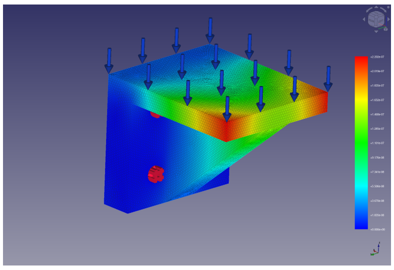
As can be seen on table 4, the Von Mises stress doesn't seem to converge, even for the refined mesh. One explanation to this is that we may be dealing here with a stress concentrator, where the stress is diverging at a corner.

6.2 Configuration B: Bracket With Stiffener

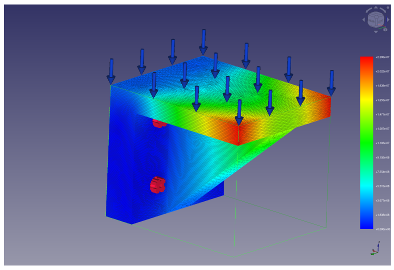
6.2.1 Displacement Results



Coarse mesh



Middle mesh



Refined mesh

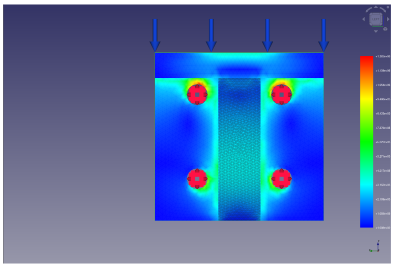
Table 5: Max. displacement magnitude for each mesh

Type of mesh	Coarse Mesh	Medium Mesh	Fine Mesh
Displacement magnitude [nm]	218,63	220,23	220,62

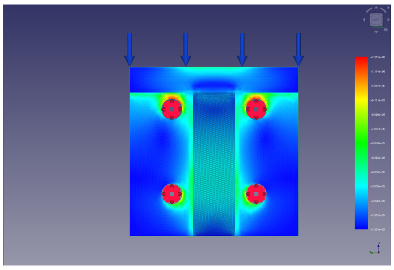
On the figures above, the effect of the plate is noticeable. Indeed, it can be seen that the deflection of the lifting face has been greatly reduced in general, and even more on the part of the lifting face that is directly supported by the stiffener. The maximum displacement is now located at the two corners of the lifting face that are not supported by the stiffener.

As can be seen in table 5, the displacement magnitude seems to have converged at around 220,62 nm. Indeed, the variation rate goes from 0,73% from the coarse mesh to the medium mesh; to 0,18% from the medium mesh to the refined mesh.

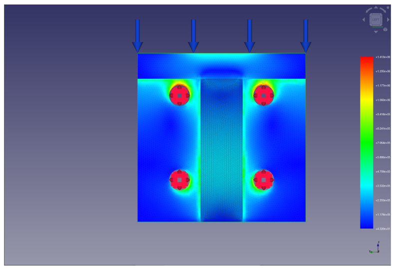
6.2.2 Von Mises Stress Results



Coarse mesh



Middle mesh



Refined mesh

On the figures above, an interesting point can be noticed: the maximum Von Mises stress has been displaced. It is not located anymore at the corner between the two plates, but rather at the top corner of the two holes closer to the lifting face.

Moreover, on table 6, it can be seen that the max. Von Mises stress is not converging, it seems to rather diverge: the more we refine the mesh, the more the max. stress value increases. This is typical of a stress concentrator, as the location of this max. stress is a corner (similarly to the case without the stiffener).

Table 6: Max. Von Mises stress for each mesh

Type of mesh	Coarse Mesh	Medium Mesh	Fine Mesh
Von Mises stress [kPa]	4926,92	5052,2	5341,73

6.3 Comparison of Results

6.3.1 Maximum Deflection Comparison

As stated previously, it can be safely assumed that the deflection converged on the refined mesh. As a reminder, the maximum deflection for both cases can be found in table 7. The addition of the stiffener thus reduced the max. displacement value by approximately 92,7 %.

Table 7: Max. converged displacement for both cases

Case	Without stiffener	With Stiffener
Max. converged displacement [nm]	3020	220,62

6.3.2 Maximum Stress Comparison

Regarding the stress, contrarily to the displacement, it cannot be assumed that the value converged on the refined mesh.

6.3.3 Mass Comparison

In order to assess how effective the stiffener is, it is necessary to compute the mass of the two parts. On table 8, the volume of the two geometries are displayed, alongside with their mass. As the mass value cannot be directly found on FREECAD, the volume value has been taken and multiplied by the density of the material (as described previously). It was found that the addition of the stiffener led to a relative mass increase of about 27,4 %.

Table 8: Volumes and masses for each of the two geometries

Type of mesh	Without stiffener	With stiffener
Volume [mm^3]	24814,6	31614,6
Mass [kg]	0,194	0,247

7. Discussion

7.1 Influence of the Stiffener

As expected, after applying the stiffener, the max. displacement was reduced from 3020 nm to 220,62 nm, which represents a decrease of about 92,7 %.

7.2 Mass versus Stiffness Trade-Off

As seen previously, a relative mass addition of about 27,4 % led to a max. displacement decrease of about 92,7%.

7.3 Limitations of the Model

In order to have the geometries the most "FEA-friendly", sharp angles were used for the geometries. However, in real life, assuming that these parts would be machined or welded, rounded corners would have been more realistic, and would have possibly decreased the influence of the stress concentrators.

7.4 Recommendations for Future Improvements

One side to explore would be to go deeper in the stress concentrator analysis, maybe by taking a mesh that is even more refined in the area surrounding the concentrators. Moreover, as mentioned just above, it could be interesting to see how the stress distribution would be modified by adding round corners, instead of sharp ones. Lastly, maybe the width of the stiffener could be reduced, in order to reduce the mass of the part, while keeping a strong max. deflection decrease.

8. Conclusion

After this FEA study, it has been found that the addition of the stiffener drastically decreased the max. displacement. The convergence study on the max. displacement was conclusive and gave us a final displacement of 3020 nm without the stiffener, and 220,62 nm with the stiffener. Regarding the Von Mises stress, however, the convergence could not be reached, the reason behind that being the possible presence of stress concentrators. One way to mitigate the influence of these concentrators could be to replace the sharp edges by round ones. In order to optimize the stiffener geometry further, it could be interesting to reduce the width of the stiffener, to save material, and then money regarding an industrial use.

References

[1] Material Database *Steel 15CrNi6 Material Properties*. Available at: `https://einsal.com/en/materials/material-database/material/1.5919#material-list` (Accessed July 2026).